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Method of fabricating non-polar and semi-polar devices using epitaxial lateral overgrowth

Abstract

A method of fabricating a semiconductor device, comprising: forming a growth restrict mask on or above a III-nitride substrate, and growing one or more island-like III-nitride semiconductor layers on the III-nitride substrate using the growth restrict mask. The III-nitride substrate has an in-plane distribution of off-angle orientations with more than 0.1 degree; and the off-angle orientations of an m-plane oriented crystalline surface plane range from about +28 degrees to about -47 degrees towards a c-plane. The island-like III-nitride semiconductor layers have at least one long side and short side, wherein the long side is perpendicular to an a-axis of the island-like III-nitride semiconductor layers. The island-like III-nitride semiconductor layers do not coalesce with neighboring island-like III-nitride semiconductor layers.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. Section 119(e) of commonly-assigned U.S. Provisional Patent Application Ser. No. 62/650,487, filed on Mar. 30, 2018, by Takeshi Kamikawa, Srinivas Gandrothula, and Hongjian Li, entitled “METHOD OF FABRICATING NON-POLAR AND SEMI-POLAR DEVICES BY USING LATERAL OVERGROWTH,” which application is incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates to a method for fabricating non-polar and semi-polar devices using epitaxial lateral overgrowth.

2. Description of the Related Art

(2) Some device manufacturers have used non-polar and semi-polar GaN substrates to produce laser diodes (LDs) and light emitting diodes (LEDs) for lighting, optical storage, etc. Non-polar and semi-polar GaN substrates are used to avoid large pyro- and piezo-electric fields, which can result in significant improvements in the emission efficiency. However, epitaxial growth of III-nitrides along non-polar and semi-polar directions is more difficult than growth along a polar c-plane direction.

(3) According to US20170092810A1, a number of pyramidal hillocks were observed on the surface of non-polar m-plane III-nitride films after epitaxial growth. Furthermore, a wavy surface appeared, which made the surface roughness worse. It is very severe problem when a laser

structure is fabricated on the surface, because this surface roughness may cause uniformity of optical gain and fluctuations of device processes. It may also decrease the yield of the device processes. Furthermore, reliability of the laser is affected by hillocks and large surface roughness. See, e.g., Applied Physics Letters 91, 191906 (2007).

(4) According to this paper, the growth conditions for obtaining a smooth surface are very narrow and with strict limitations. One problem is that surface morphology is affected by off-angle orientation, and it is well known that GaN wafers have a large in-plane distribution with regards to off-angle orientations. Consequently, the surface morphology greatly changes within the wafer. In this case, the yield is low, which is a big problem. See, e.g., Physica Status Solidi (a), Volume 214, Issue 8, 1600829 (2047).

(5) Furthermore, according to this paper, the hillock's facet dependence on leakage current under reverse-biased conditions was observed. They showed that the leakage current distribution was caused by the hillock's facet dependence on the carrier concentration and oxygen concentration. The leakage current distribution is a problem when making devices, such as LEDs, LDs, power devices such as Schottky barrier diodes (SBDs), or metal-oxide-semiconductor field-effect-transistors (MOSFETs).

(6) Another problem is growing epi-layers on a growth restrict mask using a non-polar and semi-polar III-nitride substrate results in the epi-layers coalescing at the center of an opening area in the mask. In this case, the island-like III-nitride layers have a concave region or depression at the near-center of the epi-layers. Under different growth conditions, there are cases where the island-like III-nitride layers have a convex region or elevation at the near-center of the epi-layers. Both cases present problems when attempting to obtain a flat surface of the epi-layers.

(7) Thus, there is a need in the art for improved methods for flattening the epilayer on the non-polar and semi-polar substrate using epitaxial lateral overgrowth (ELO). The present invention satisfies this need.

SUMMARY OF THE INVENTION

(8) To overcome the limitations in the prior art described above, and to overcome other limitations that will become apparent upon reading and understanding this specification, the present invention discloses a method of fabricating a semiconductor device, comprising: forming a growth restrict mask on or above a III-nitride substrate, wherein: the III-nitride substrate has an in-plane distribution of off-angle orientations with more than 0.1 degrees; the off-angle orientations of an m-plane oriented crystalline surface plane ranges from about +28 degrees to about -47 degrees towards a c-plane; and growing one or more island-like III-nitride semiconductor layers on the III-nitride substrate using the growth restrict mask. The island-like semiconductor layers are then removed from the III-nitride substrate.

(9) The resulting island-like III-nitride semiconductor layer has a just-orientation and an off-angle orientation of an m-plane oriented crystalline surface plane, wherein: the off-angle orientation of the m-plane oriented crystalline surface plane ranges from about +28 degrees to about -47 degrees towards a c-plane; the island-like III-nitride semiconductor layers have at least one long side and short side, wherein the long side is perpendicular to an a-axis of the island-like III-nitride semiconductor layers; and the island-like III-nitride semiconductor layers do not coalesce with neighboring island-like III-nitride semiconductor layers.

(10) The island-like III-nitride semiconductor layers have an emitting region, wherein the emitting region is at least 1 μm from an edge of a layer bending region of the island-like III-nitride semiconductor layers. The emitting region is more than 5 μm from an edge of a top surface of the island-like III-nitride semiconductor layers.

(11) The island-like III-nitride semiconductor layers have an edge growth region with a height less than 0.2 μm , wherein the edge growth region has a width less than 5 μm .

(12) The island-like III-nitride semiconductor layers also have a separate region where an n-electrode is formed.

(13) The present invention also discloses a method for growing a flat III-nitride epi-layer on a semi-polar and non-polar III-nitride substrate using ELO, thereby reducing or eliminating pyramidal hillocks, wavy surfaces, and concave or convex regions.

(14) In one aspect, it has been realized that the edge of the substrate causes the roughness of the surface to deteriorate. Therefore, a region of growth (e.g., an opening area in a growth restrict mask) needs to be isolated from the edge of the substrate. In this invention, the region of growth has been isolated by the growth restrict mask.

(15) Furthermore, the substrate may have pits and defects on its surface, which results in a rough surface after epitaxial growth using metal-organic chemical vapor deposition (MOCVD). This has been well-known for some time. One pit may affect the surface roughness in a large area. However, separating each of the regions of growth can prevent worse surface roughness in adjacent regions. In semi-polar and non-polar epitaxial growth, it is important for the regions of growth to be isolated from the edge of substrate and to not affect a large area.

(16) In other words, the region of growth is surrounded with the growth restrict mask, which isolates the region of growth. For example, the growth restrict mask may be a dielectric film or metals, such as SiO₂, SiN, HfO₂, Al₂O₃, MgF, W, Mo, etc. Furthermore, the growth restrict mask may use multiple layers selected from these materials.

(17) A III-nitride layer grown using the ELO method is called an ELO III-nitride layer. When the ELO III-nitride layer is grown using MOCVD or HVPE, a carrier gas flow includes a Hydrogen gas.

(18) If the ELO III-nitride layers coalesce with each other, it causes fluctuations in the surface roughness. Furthermore, the region of coalescence causes many stacking faults and misfit dislocations.

(19) Therefore, in this invention, the growth of the ELO III-nitride layer is stopped before the adjacent island-like layers coalesce with each other. Then, the flat region of the ELO III-nitride layer is substantially free from hillocks. Thereafter, III-nitride semiconductor device layers are grown on the ELO III-nitride layer, and the III-nitride semiconductor device layers are removed from the III-nitride substrate. These are separated from each other because MOCVD growth is stopped before they coalesce. The method obtains flat surface layers and device layers with high yield.

(20) The present invention can use a III-nitride substrate or a hetero-substrate. More preferably, the present invention uses the III-nitride substrate, in order to obtain high quality III-nitride-based semiconductor layers and to avoid bowing or curvature of the substrate during epitaxial growth. The present invention also obtains low defect density devices.

(21) As long as it enables growth of a III-nitride based semiconductor layer through a growth restrict mask, any III-nitride based substrate or hetero-substrate, such as sapphire, SiC, LiAlO₂, Si, etc., can be used.

(22) Moreover, III-nitride based semiconductor layers and a III-nitride based substrate may also include Al, In, and B, as well as other impurities such as Mg, Si, O, C, H, etc.

(23) The ELO III-nitride layers are epitaxially grown on the III-nitride substrate at an opening area and/or through an intermediate layer at the opening area. The quality of the ELO III-nitride layer and the III-nitride semiconductor device layers are extremely high, and a device comprised of the island-like III-nitride based semiconductor layers is of extremely high quality. Furthermore, the island-like III-nitride semiconductor layers have a flat surface morphology without hillocks, which usually appear on an epi-layer surface after growing the epi-layer on a non-polar and semi-polar substrate.

(24) In MOCVD or HVPE, it is better to use carrier gases that include Hydrogen gas. Hydrogen plays a crucial role in the beginning of the growth. According to the prior art, it has been shown that a Hydrogen carrier gas makes the surface morphology worse during GaN layer growth with MOCVD. However, when the region of growth is surrounded with a growth restrict mask, the

situation changes.

(25) Carrier gases that include Hydrogen effectively etch an edge-side of the epi-layers in an opening area. This effect prevents the epi-layers from beginning growth at both sides of the opening area.

(26) If carrier gases do not include Hydrogen, there are a lot of cores of the epi-layers at both sides of an opening area. With further growth, cores of the epi-layers would coalesce at or near the center of opening area. Therefore, the ELO III-nitride layer has a depressed region. This growth has been happened on a variety of non-polar and semi-polar planes.

(27) On the other hand, carrier gases that include Hydrogen gas may be used, where the cores of the epi-layers form at or near the center of opening area in the early stage of growth. In this case, the ELO III-nitride layer does not have any depressed regions or pyramidal hillocks, and a very flat and smooth surface roughness can be obtained.

(28) For above reasons, when the ELO III-nitride layer is grown, it would better to use a carrier gas including Hydrogen at least during the beginning of growth. Moreover, the carrier gas can be only Hydrogen gas or a mix of Hydrogen gas and Nitrogen gas.

(29) In this invention, it is also possible to remove the island-like layers from the substrate. It is difficult for a homo-epitaxial layer to be removed from a substrate, because there is no hetero-interface between the homo-epitaxial layer and the substrate's surface. However, using this invention, the homo-epitaxial island-like layers can be removed from the substrate in a fast and easy manner.

(30) For example, the method of removing substrate may use a growth restrict mask, which may be a dielectric film or metals, such as SiO_2 , SiN , HfO_2 , Al_2O_3 , MgF , etc. The interface between the growth restrict mask and any subsequent III-nitride layers grown by ELO on the mask has a weak bonding strength. The bonding area (opening area width) is controlled to be more or less than the chip size. Besides, the ELO method provides a cleaving point and cleaving interface. Thus, it is easy to peel off the homo-epitaxial layers from the substrate.

(31) Furthermore, these methods use cleavage with the m-plane. The m-plane is the easiest plane among GaN planes to cleave. This method also can determine the cleaving point in an easy manner. For example, the cleaving point may be an edge of the growth restrict mask. One improvement is to dissolve the mask using a hydrofluoric acid (HF), buffered HF (BHF), or another etchant, before removing the substrate.

(32) Thereafter, the wafer is bonded to a support substrate using a low temperature melted metal and/or solder, wherein the metal is dissolved by the etchant. The bonding parts are the III-nitride layers on the III-nitride substrate. It is possible to utilize support substrates that have a thermal expansion different from the III-nitride substrate. Both substrates are heated or cooled after bonding. Stress is applied to the III-nitride layers that are bonded to the support substrate due to differences in thermal expansion. This stress is applied to the portion of the III-nitride layers that are between the III-nitride layer and the substrate. The cleave starts from a cleaving point that is an edge of the growth restrict mask. In the end, the cleave reaches an opposite side of the cleaving point. However, a trigger is needed in order to start the cleave.

(33) As in the above case, the cleave may be triggered using the stress from the difference in thermal expansion. However, it may be unnecessary to use such stress. For example, the cleave may be triggered using ultra-sonic waves, etc. If mechanical removal, such as ultra-sonic cleaving, can be used, the substrate can be removed quickly and with very weak stress, due to the cleaving of the m-plane. Furthermore, the cleaving point is a wedge shape, which makes the determination of the cleaving point easy. Moreover, the shape of the cleaving point is important to achieve a high yield.

(34) Using these methods, device layers can be easily removed from the III-nitride-based substrates and wafers, including wafers of large size, e.g., over 2 inches.

(35) In addition, the island-like III-nitride based semiconductor layers do not coalesce with each

other, and internal strain is released, which avoids any occurrences of cracks.

(36) As a main purpose of this present invention, even if the III-nitride based substrate or hetero-substrate has a large in-plane distribution of off-angle orientations, a smooth surface is obtained with a large area without hillocks or wavy roughness.

(37) In the present invention, the island-like III-nitride layer need not be removed from the substrate. If the island-like III-nitride layer is removed from the substrate, the process described above can be used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Referring now to the drawings in which like reference numbers represent corresponding parts throughout:

(2) FIG. 1 is a schematic of a structure comprised of a substrate with various layers deposited thereon, according to an embodiment of the present invention.

(3) FIGS. 2(a), 2(b), 2(c), 2(d), 2(e) and 2(f) are schematics illustrating fabrication of the structure, according to an embodiment of the present invention.

(4) FIGS. 3(a) and 3(b) are schematics illustrating growth of semiconductor layers on a growth restrict mask, according to an embodiment of the present invention.

(5) FIGS. 4(a) and 4(b) are schematics illustrating a growth restrict mask, according to an embodiment of the present invention.

(6) FIG. 5 is a photograph illustrating experimental results, according to an embodiment of the present invention.

(7) FIGS. 6(a), 6(b), 6(c), 6(d), 6(e), 6(f), 6(g), 6(h) and 6(i) are photographs illustrating experimental results, according to an embodiment of the present invention.

(8) FIGS. 7(a), 7(b) and 7(c) are schematics illustrating a growth restrict mask, according to an embodiment of the present invention.

(9) FIGS. 8(a) and 8(b) are schematics illustrating a device structure and its fabrication, according to an embodiment of the present invention.

(10) FIGS. 9(a) and 9(b) are schematics illustrating a process of structure removal and FIG. 9(c) is a photograph of the removed structure, according to an embodiment of the present invention.

(11) FIGS. 10(a), 10(b), 10(c) and 10(d) are photographs illustrating experimental results, according to an embodiment of the present invention.

(12) FIGS. 11(a) and 11(b) are schematics illustrating crystal orientations, according to an embodiment of the present invention.

(13) FIG. 12 is a schematic illustrating a device structure, according to an embodiment of the present invention.

(14) FIG. 13 is a series of photographs illustrating experimental results, according to the present invention.

(15) FIGS. 14(a) and 14(b) are schematics and photographs illustrating experimental results, according to the present invention.

(16) FIGS. 15(a) and 15(b) are schematics illustrating semiconductor growth structures, and FIG. 15(c) is a photograph of the structure of FIG. 15(b), according to an embodiment of the present invention.

(17) FIG. 16 is a schematic and photograph illustrating experimental results, according to an embodiment of the present invention.

(18) FIGS. 17(a), 17(b) and 17(c) are schematics illustrating growth of semiconductor layers on a growth restrict mask, according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(19) In the following description of the preferred embodiment, reference is made to a specific embodiment in which the invention may be practiced. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention.

(20) Overview

(21) The present invention describes a method of fabricating a semiconductor device, including obtaining a smooth and flat surface for an epitaxial III-nitride layer grown on a non-polar or semi-polar III-nitride substrate, and then manufacturing a semiconductor device on that surface.

(22) In one embodiment, the method comprises the steps of: forming a growth restrict mask with a plurality of striped opening areas directly or indirectly upon a substrate, which may be a III-nitride substrate or a hetero-substrate, wherein: the substrate has an in-plane distribution of off-angle orientations with more than 0.1 degrees; the off-angle orientations of an m-plane oriented crystalline surface plane range from about +28 degrees to about -47 degrees towards a c-plane; and the opening areas have long sides and short sides, the long sides are perpendicular to an a-axis direction; and growing one or more island-like III-nitride semiconductor layers on the substrate using the growth restrict mask, wherein: the island-like III-nitride based semiconductor layers are grown in a Hydrogen atmosphere; and the growth extends in a direction parallel to the striped opening areas of the growth restrict mask, wherein the island-like III-nitride based semiconductor layers do not coalesce; and removing the island-like III-nitride semiconductor layers from the III-nitride substrate.

(23) The resulting island-like III-nitride semiconductor layers have a just-orientation and an off-angle orientation of an m-plane oriented crystalline surface plane, wherein: the off-angle orientation of the m-plane oriented crystalline surface plane ranges from about +28 degrees to about -47 degrees towards a c-plane; the island-like III-nitride semiconductor layers have at least one long side and short side, wherein the long side is perpendicular to an a-axis of the island-like III-nitride semiconductor layers; and the island-like III-nitride semiconductor layers do not coalesce with neighboring island-like III-nitride semiconductor layers.

(24) The island-like III-nitride semiconductor layers have an emitting region, wherein the emitting region is at least 1 μm from an edge of a layer bending region of the island-like III-nitride semiconductor layers, and the emitting region is more than 5 μm from an edge of a top surface of the island-like III-nitride semiconductor layers.

(25) The island-like III-nitride semiconductor layers have an edge growth region with a height less than 0.2 μm and a width less than 5 μm .

(26) The island-like III-nitride semiconductor layers also have a separate region where an n-electrode is formed.

(27) Finally, a device fabricated using the island-like III-nitride semiconductor layers may comprise a light-emitting diode (LED), laser diode (LD), Schottky barrier diode (SBD), or metal-oxide-semiconductor field-effect-transistor (MOSFET), which is processed on a flat surface region and/or the opening areas.

(28) Semiconductor Structure and Fabrication Method

(29) FIG. 1 is a schematic that illustrates a semiconductor structure according to an embodiment of the present invention, wherein the structure includes a substrate **101**, growth restrict mask **102**, opening areas **103**, no-growth region **104**, ELO III-nitride layer **105**, III-nitride semiconductor device layers **106**, flat surface region **107**, layer bending region **108**, and island-like III-nitride semiconductor layers **109** (which are formed on the flat surface region **107** and layer bending region **108**) that may include a light emitting region **110**.

(30) The method for fabricating the semiconductor structure includes the following steps:

(31) 1. ELO+III-Nitride Semiconductor Layers.

(32) As shown in FIG. 2(a), III-nitride layers **105**, such as GaN layers, are grown by ELO on a substrate **101** patterned with an SiO₂ growth restrict mask **102**.

- (33) The substrate **101** may comprise, for example, a GaN substrate, AlN substrate, m-plane sapphire substrate, etc. In one embodiment, a GaN substrate **101** is used that has a growth surface that is m-plane with an off-angle from -47 degrees to $+47$ degrees towards a c-plane (0001).
- (34) The patterned SiO.sub.2 of the growth restrict mask **102** has a width of $20\text{ }\mu\text{m}$ and an interval of $150\text{ }\mu\text{m}$, wherein the SiO.sub.2 stripes of the growth restrict mask **102** are along the $\langle 0001 \rangle$ axis. The ELO III-nitride layers **105** do not coalesce on top of the SiO.sub.2 of the growth restrict mask **102**. Epi-layers including III-nitride semiconductor device layers **106** are then grown, having flat surface region **107** and layer bending region **108**, resulting in island-like III-nitride semiconductor layers **109**, which may include GaN layers, AlGaIn layers, AlInGaIn layers, etc.
- (35) A ridge process may then be carried out on the island-like III-nitride semiconductor layers **109** on the flat surface region **107** to form LD device. Alternatively, an LED device or other device may be fabricated.
- (36) 2 Dissolving the Growth Restrict Mask by Wet Etching.
- (37) As shown in FIG. 2(b), the SiO.sub.2 of the growth restrict mask **102** is dissolved using a chemical solution, such as hydrofluoride (HF), buffered HF (BHF), etc., or otherwise removed. This makes it easier to cleave devices from the GaN substrate **101**. This process is best performed before removing the epi-layers from the substrate **101**.
- (38) 3. TCO p-Pad Deposition+Ridge Process.
- (39) As shown in FIG. 2(c), a ZrO.sub.2 layer **201** is patterned, and top-side p-contacts **202**, such as TCO (Transparent Conductive Oxide), are deposited on the devices, followed by a p-pad **203**. In addition, p-AlGaIn cladding layers and electrodes can be used, but this invention is not limited to those structures.
- (40) 4. Bonding the Support Substrate.
- (41) As shown in FIG. 2(d), the devices are flip-chip bonded to a carrier wafer (Si, Cu, Cu—W, etc.) as a support substrate **204** using metal-metal bonding or soldering techniques.
- (42) 5. Heat the Support Substrate.
- (43) As shown in FIG. 2(c), the support substrate **204** is heated to remove the devices from the substrate **101** at a cleaving point **205** and across a cleaving surface **206**. Alternatively, the support substrate **204** may be cooled to accomplish this function.
- (44) 6. N-Electrode Deposition.
- (45) As shown in FIG. 2(f), back-side n-contacts **207**, such as TCO, Ti/Al, Ti/Au, Hf/Al/Mo/Au, etc., are deposited on to the devices. However, the n-contacts **207** are not limited to those materials.
- (46) 6. Chip Scribing.
- (47) Chip scribing may be performed to separate the devices, as shown in FIGS. 3(a)-3(b), wherein FIG. 3(a) shows a top view of the flat surface regions **107**, layer bending regions **108**, and island-like III-nitride based semiconductor layers **109**, separated by no-growth regions **104**, and FIG. 3(b) is an enlarged portion of FIG. 3(a) that shows additional features of the LD devices, including ridge stripe structures **301**, etched mirror regions **302** and chip scribe lines **303**.

Definitions of Terms

(48) III-Nitride Based Substrate

(49) As long as a III-nitride based substrate **101** enables growth of a III-nitride based semiconductor layer through a growth restrict mask **102**, any III-nitride substrate **101** that is sliced on a $\{1-100\}$, $\{20-21\}$, $\{20-2-1\}$, $\{10-11\}$, $\{10-1-1\}$ plane, etc., or other plane, such as from a bulk GaN or AlN crystal, can be used. The bulk GaN substrate off-angle orientation is from the m-plane towards the c-plane at 0 degrees to ± 47 degrees.

(50) The III-nitride based substrate **101** and bulk GaN may include Al, In, B, etc.

(51) III-Nitride Based Semiconductor Layers

(52) The III-nitride based semiconductor layers include the ELO III-nitride layer **105**, the III-nitride semiconductor device layers **106** and the island-like III-nitride based semiconductor layers **109**.

These III-nitride-based semiconductor layers can include In, Al and/or B, as well as other

impurities, such as Mg, Si, Zn, O, C, H, etc.

(53) The ELO III-nitride layer **105** generally comprises a GaN layer, although other III-nitride layers may be used.

(54) The III-nitride semiconductor device layers **106** generally comprise two or more layers, including at least one layer among an n-type layer, an undoped layer and a p-type layer. The III-nitride semiconductor device layers **106** may comprise one or more GaN layers, AlGaN layers, InGaN layers, AlGaInN layers, etc.

(55) The island-like III-nitride based semiconductor layer **109** are typically formed with sides along an (1-10a) plane (where a is an arbitrary integer), (11-2b) plane (where b is an arbitrary integer), or planes crystallographically equivalent to these, or the sides of the island-like III-nitride semiconductor layers **109** include the (1-10a) plane (where a is an arbitrary integer).

(56) The distance between the island-like III-nitride semiconductor layers **109** adjacent to each other is generally 30 μm or less, and preferably 10 μm or less, but is not limited to these values. The distance between the island-like III-nitride semiconductor layers **109** is preferably the width of a no growth region **104**.

(57) As shown FIGS. **4(a)** and **4(b)**, the island-like III-nitride semiconductor layer **109** has short sides **401** and long sides **402**, wherein the long sides **402** are perpendicular to the a-axis.

(58) In various embodiments, the island-like III-nitride semiconductor layers **109** may be used to fabricate a light-emitting diode, a laser diode, a Schottky diode, a photodiode, a transistor, etc., but is not limited to these devices. This invention is particularly useful for micro-LEDs and LDs, such as edge-emitting lasers (EELS) and vertical cavity surface-emitting lasers (VCSELs).

(59) The number and placement of electrodes depend on the type of semiconductor device, and typically are disposed at predetermined portions.

(60) Growth Restrict Mask

(61) The growth restrict mask **102** comprises a dielectric layer, such as SiO₂, SiN, SiON, Al₂O₃, AlN, AlON, MgF, or a refractory metal or precious metal, such as W, Mo, Ta, Nb, Rh, Ir, Ru, Os, Pt, etc. The growth restrict mask **102** may be a laminate structure selected from the above materials. It also can be a multiple-stacking layer structure, which is chosen from the above materials.

(62) In one embodiment, the thickness of the growth restrict mask **102** is about 0.05-3 μm . The width of the mask is preferably larger than 20 μm , and more preferably, the width is larger than 40 μm .

(63) The Effect of Hydrogen Etching

(64) In the present invention, the carrier gas can include Hydrogen gas. The Hydrogen gas has the effect of etching GaN layers or reducing their growth rate. The width of the growth restrict mask **102** affects these effects. A GaN layer is not grown on the growth restrict mask **102**, so that area's consumption of Hydrogen for etching the GaN layer is very low.

(65) Therefore, the numbers of Hydrogen atoms that reach the edge of the opening area **103** increase, so that the edge of the opening area **103** is affected by Hydrogen etching strongly. On the other hand, the number of Hydrogen atoms that reach the center of the opening area **103** decreases as compared to the edge.

(66) This effect can be seen in FIG. **5**. To simplify, the patterned substrate **101** has a wide open area of about 100 μm . As shown in FIG. **5**, the edge area **501** of the opening area **103** is thinner than its center area, which shows the effect of Hydrogen etching.

(67) This effect also affects the growth of the layer at the beginning, as shown in FIGS. **6(a)**-**6(i)**. For example, as shown in FIG. **6(b)**, the effect of Hydrogen etching makes the core **601** form at the center of the opening area **103**, which avoids forming a depressed region at the center of the ELO III-nitride layer **105**.

(68) For the reasons mentioned above, the wider the growth restrict mask **102**, the stronger the effects of Hydrogen etching at the edge of the opening area **103**. Therefore, the width of the mask

102 is preferably larger than 20 μm , and more preferably, the width is larger than 40 μm . However, the width of the growth restrict mask **102** is preferably under 180 μm , in case there is debris on the growth restrict mask **102**.

(69) Furthermore, the width of the opening area **103** is preferably more than 2 μm . If the width of the opening area **103** is less than 2 μm , it is difficult to grow GaN layers at the opening area **103** with a Hydrogen carrier gas condition, due to the effects of Hydrogen etching.

(70) The Direction of the Growth Restrict Mask

(71) In one example, the growth restrict mask **102** comprises a plurality of striped opening areas **103**, shown as **103** in FIGS. 7(a)-7(c). The striped opening areas **103** are arranged in a first direction parallel to the 11-20 direction of the III-nitride based semiconductor layers and a second direction parallel to the 0001 direction of the III-nitride based semiconductor layers, periodically at a first interval and a second interval, respectively, and extend in the second direction.

(72) The width of the striped opening areas **103** is typically constant in the second direction, but may be changed in the second direction as necessary.

(73) The growth restrict mask **102** comprises a plurality of striped opening areas **103** which are arranged in the first direction parallel to the 11-20 direction of the ELO III-nitride layer **105** periodically and extend in the second direction parallel to the 1-100 direction of the ELO III-nitride layer **105**; and a plurality of striped opening areas **103** which are arranged periodically in the first direction at the same interval as the striped opening areas **103** and shifted by a half of the interval with respect to the striped opening areas **103** and extend in the second direction in a manner, such that the plurality of striped opening areas **103** overlap with end portions of the striped opening areas **103** for the predetermined distance in the second direction. The width of these striped opening areas **103** is typically constant in the second direction but, as necessary, may be changed in the second direction.

(74) As shown in FIG. 7(c), in this case, the substrate **101** does not have an overlap area. The pitch of *c* is the distance between islands in the direction parallel to the 1-100 direction. In this case, it is easy to obtain a smooth surface without hillocks.

(75) Flat Surface Region

(76) The flat surface region **107** is between layer bending regions **108**. Furthermore, the flat surface region **107** is on the growth restrict mask **102**.

(77) Fabrication of the semiconductor device is mainly performed on the flat surface region **107**. The width of the flat surface region **107** is preferably at least 5 μm , and more preferably is 10 μm or more. The flat surface region **107** has a high uniformity for the thickness of each semiconductor layer.

(78) Layer Bending Region

(79) FIGS. 8(a)-8(b) illustrate the layer bending region **108**, as well as a bended active region **801** that may remain in the device. The definition of a layer bending region **108** is the region outside of the bended active region **801** including the bended active region **801**.

(80) The fabrication of the device may be partially performed on the layer bending region **108**. More preferably, the layers at the bending layer region **108** are removed by etching. For example, it is better that at least a part of an active layer in the layer bending region **108** is removed by using an etching process, such as dry etching or wet etching.

(81) If a non-polar or semi-polar substrate **101** is used, the island-like III-nitride semiconductor layers **109** have two or three facets **802**, **803**, **804** at one side. In the case of three facets, the first facet **802** is a main area to form a ridge structure, while the second facet **803** and third facet **804** are included the layer bending region **108**.

(82) If the layer bending region **108** that includes an active layer remains in an LED device, a portion of the emitted light from the active layer is reabsorbed. As a result, it is preferable to remove at least a part of the active layer in the layer bending region **108** by etching.

(83) If the layer bending region **108** that includes an active layer remains in an LD device, the laser

mode may be affected by the layer bending region **108** due to a low refractive index (e.g., an InGaN layer). As a result, it is preferable to remove at least a part of the active layer in the layer bending region **108** by etching. More preferably, two etchings may be performed, wherein a first etching removes the active layer in the second facet **803** region before removing the epi-layers from the substrate **101** and a second etching removes the active layers in the third facet **804** region after removing the epi-layers from the substrate **101**. If the layer bending region **108** remains in the LD device, the edge of the ridge stripe structure should be at least 1 μm or more from the edge of the layer bending region **108**.

(84) The emitting region is a current injection region. In the LD case, the emitting region is a ridge structure. In the LED case, the emitting region is the region where the p-contact electrode is formed. In both the LD and LED cases, the edge of the emitting region should be at least 1 μm or more from the edge of the layer bending region **108** and more preferably 5 μm .

(85) From another point of view, the epitaxial layer of the flat surface region **107**, except for the opening area **103**, is of a lesser defect density than the epitaxial layer of the opening area **103**. Therefore, it is more preferable for the ridge stripe structure to be formed on the flat surface region **107** including its wings.

(86) First and Second Support Substrates

(87) The method for manufacturing the semiconductor device, as necessary, may further comprise a step of bonding a first support substrate to the exposed surface side of the island-like III-nitride semiconductor layers **109** and a first support substrate to the exposed surface side of the III-nitride substrate **101**, before peeling the island-like III-nitride semiconductor layers **109** from the III-nitride substrate **101**. The first and second support substrates may be comprised of elemental semiconductor, compound semiconductor, metal, alloy, nitride-based ceramics, oxide-based ceramics, diamond, carbon, plastic, etc., and may comprise a single layer structure or a multilayer structure made of these materials. A metal, such as solder, etc., or an organic adhesive, may be used for the bonding of the first and second support substrates, and is selected as necessary.

(88) Support Film

(89) FIGS. **9(a)-9(b)** illustrate the use of a support film **901** in removing devices from the substrate **101**. The support film **901** may be an adhesive tape, UV tape, polyimide tape, etc., but is not limited to those materials.

(90) In one embodiment, the support film **901** comprises a layer of polyimide tape (20-50 μm) that is roll-applied to the surface of the metal layer of the p-pad **203**. A fracture occurs at the cleaving point **205** upon gently pulling the tape **901** away from the surface of the substrate **101**, and along the cleaving surface **206**. An image of a device bar removed in this manner is shown in FIG. **9(c)**.

(91) Also, to prevent excessive bending of the thin film after spalling, the outer portions of the tape **901** may be used to secure the sample to a frame.

(92) Fabrication Method

(93) The method of manufacturing the semiconductor device may further comprise a step of forming the growth restrict mask **102** on the III-nitride substrate **101**.

(94) For example, m-plane (1-100) freestanding GaN substrates **101** may be used with a growth restrict mask **102** of SiO₂. As shown in FIGS. **7(a)-7(c)**, opening areas **103** are in the direction parallel to $\langle 1-100 \rangle$. If semi-polar (20-21), (20-2-1) substrates **101** are used, the opening areas **103** are in the direction parallel to $[-1014]$, $[10-14]$, respectively. The opening areas **103** have a width b , which is 2 μm -180 μm , and more preferably, 4 μm -50 μm , which makes it is easy to obtain to a smooth surface.

(95) Metal-organic chemical vapor deposition (MOCVD) is used for the epitaxial growth.

Trimethylgallium (TMGa), trimethylindium (TMIn) and triethylaluminium (TMAI) are used as III elements source. Ammonia (NH₃) is used as the raw gas to supply Nitrogen. Hydrogen (H₂) and Nitrogen (N₂) are used as carrier gas of III elements sources. It is important to include Hydrogen in the carrier gas to obtain a smooth surface for the epi-layer.

(96) Saline and Bis(cyclopentadienyl)magnesium (Cp.sub.2Mg) is used as the n-type and p-type dopants. The pressure is setting to be 50 to 760 Torr. The GaN growth temperature ranges from 1050 to 1250° C.

(97) After about 2-8 hours growth time, island-like III-nitride semiconductor layers **109** have the following dimensions: Thickness: 20-60 μm; and Bar width: 40-150 μm (where the bar width is the width of the island-like III-nitride semiconductor layers **109**).

(98) The method may include bonding the first support substrate to the upper surface side of the III-nitride based semiconductor layers after growing the III-nitride based semiconductor layers upon the substrate **101**, and then peeling the first support substrate and the III-nitride based semiconductor layers from the III-nitride based substrate **101**.

(99) In addition, the method may further comprise a step of forming one or more electrodes on the surface of the island-like III-nitride semiconductor layer **109** that is exposed after peeling the first support substrate and the island-like III-nitride semiconductor layers **109** from the substrate **101**.

(100) As necessary, the method of manufacturing the semiconductor device may further comprise a step of forming one or more electrodes on the upper surface of the island-like III-nitride semiconductor layers **109** after growing the island-like III-nitride semiconductor layers **109** upon the substrate **101**. The n-electrodes may be formed after III-nitride based semiconductor layers have been removed using a cleaving technique.

(101) The method may further comprise a step of removing, by wet etchant, at least a portion of, or preferably almost all of, or most preferably all of, the growth restrict mask **102**. However, this process is not always necessary to remove the substrate **101**. Also as necessary, a conductor thin film or a conductor line may be formed on the major surface of the first support substrate on the side bonded with the III-nitride based semiconductor layers.

(102) According to the present invention, the crystallinity of the island-like III-nitride semiconductor layers **109** laterally growing upon the growth restrict mask **102** from a striped opening area **103** of the growth restrict mask **102** is very high, and III-nitride based semiconductor layers made of high quality semiconductor crystal can be obtained.

(103) Furthermore, two advantages may be obtained using a III-nitride based substrate **101**. One advantage is that a high-quality island-like III-nitride semiconductor layer **109** can be obtained, such as with a very low defects density, in comparison to using a sapphire substrate **101**. Another advantage, by using a similar or same material for both the epilayer and the substrate **101**, is that it can reduce the strain in the epitaxial layer. Also, thanks to a similar or same thermal expansion, the method can reduce the amount of bending of the substrate **101** during epitaxial growth. The effect, as above, is that the production yield can be high in order to improve the uniformity of temperature. But, a hetero-substrate **101** can also be used, such as sapphire(m-plane), LiAlO.sub.2, SiC, Si, etc. More preferably, free-standing III-nitride substrates **101** can be used, due to above reasons. If a hetero-substrate **101** is used, it is easy to remove due to weak bonding strength at the cleaving point.

(104) Consequently, the present invention discloses: a substrate **101** comprised of a III-nitride based semiconductor; a growth restrict mask **102** with one or more striped opening areas **103** disposed directly or indirectly upon the substrate **101**; and one or more island-like III-nitride semiconductor layers **109** grown upon the substrate **101** using the growth restrict mask **102** in the (1-100) plane orientation, wherein the striped opening areas **103** of the growth restrict mask **102** have long sides and short sides, wherein the long sides are in a direction perpendicular to the a-axis direction of the island-like III-nitride semiconductor layers **109**, as shown in FIGS. 7(a), 7(b) and 7(c).

(105) In one embodiment, the growth restrict mask **102** is deposited by sputter or electron beam evaporation or PECVD (plasma-enhanced chemical vapor deposition); but is not limited to those methods.

(106) Also, when a plurality of island-like III-nitride semiconductor layers **109** are grown, these

layers **109** are separated each other, that is, is formed in isolation, so tensile stress or compressive stress generated in each island-like III-nitride semiconductor layers **109** is limited within the island-like III-nitride based semiconductor layers **109**, and the effect of the tensile stress or compressive stress does not fall upon the other III-nitride based semiconductor layers.

(107) Also, as the growth restrict mask **102** and the ELO III-nitride layer **105** are not bonded chemically, the stress in the ELO III-nitride layer **105** can be relaxed by a slide caused at the interface between the growth restrict mask **102** and the ELO III-nitride layer **105**.

(108) Also, the existence of gaps between each of the island-like III-nitride semiconductor layers **109**, as shown by the no-growth region **104** in FIGS. **1** and **4(a)**-**4(b)**, results in the substrate **101** having rows of a plurality of island-like III-nitride semiconductor layers **109**, which has flexibility, and therefore, is easily deformed when external force is applied and can be bended.

(109) Therefore, even if a slight warpage, curvature, or deformation occurs in the substrate **101**, this can be easily corrected by a small external force, to avoid the occurrence of cracks. As a result, the handling of substrates **101** by vacuum chucking is possible, which makes the manufacturing process of the semiconductor devices more easily carried out.

(110) As explained, island-like III-nitride semiconductor layers **109** made of high quality semiconductor crystal can be grown by suppressing the curvature of the substrate **101**, and further, even when the III-nitride based semiconductor layer is very thick, the occurrences of cracks, etc., can be suppressed, and thereby a large area semiconductor device can be easily realized.

First Embodiment

(111) A III-nitride based semiconductor device and a method for manufacturing thereof according to a first embodiment are explained.

(112) In the first embodiment, the substrate **101** is first provided, and the growth restrict mask **102** that has a plurality of striped opening areas **103** is formed on the substrate **101**. In this embodiment, the substrate **101** is made of a III-nitride semiconductor, such as GaN.

(113) Some substrates **101** may be prepared with a different off-angle orientation. FIG. **6(d)** has four SEM (Scanning Electron Microscopy) images of the surface of the island-like III-nitride semiconductor layer **109** on substrates **101** with different off-angle orientations, labeled as mis-orientations “(towards c-axis, a-axis),” where the carrier gas is H.sub.2. The off-angle orientations range up to -1 degrees from the m-plane towards the c-axis (i.e., c-plane), and 0 degrees (i.e., less than 0.02 degrees) towards the a-axis (i.e., a-plane). In this embodiment, the substrate **101** has a just-orientation of less than 0.2 degrees.

(114) Making a Patterned Substrate

(115) The thickness of the III-nitride based semiconductor layers to be grown upon the GaN substrates is about 5 to 80 μm , for example, but is not limited to these values. As described herein, the thickness of the III-nitride based semiconductor layers is measured from the surface of growth restrict mask **102** to the upper surface of the III-nitride based semiconductor layers.

(116) The growth restrict mask **102** can be formed from an insulator film, for example, an SiO.sub.2 film deposited upon the substrate **101** by plasma chemical vapor deposition (CVD), sputtering, ion beam deposition (IBD), etc., wherein the SiO.sub.2 film is then patterned by photolithography using a predetermined photo mask and etching. The thickness of the SiO.sub.2 film in this embodiment is 0.3 μm , but is not limited to that value.

(117) Using the growth restrict mask **102**, the ELO III-nitride layer **105** is grown in an island-like shape in the (1-100) plane orientation by a vapor-phase deposition method, for example, a metalorganic chemical vapor deposition (MOCVD) method. In this case, the surface of the substrate **101** is exposed in the opening areas **103**, and the ELO III-nitride layer **105** is selectively grown thereon, and is continuously laterally grown upon the growth restrict mask **102**. The growth is stopped before the ELO III-nitride layer **105** coalesces with an adjacent ELO III-nitride layer **105**.

(118) The thickness of ELO III-nitride layer **105** is important, because it determines the width of

the flat surface region **107**. Preferably, the width of the flat surface region **107** is 20 μm or more.

(119) The thickness of ELO III-nitride layer **105** is preferably as thin as possible. This is to reduce the process time and to etch the opening area **103** easily.

(120) The growth ratio of ELO is the ratio of the growth rate of the lateral direction parallel to the 11-20 axis of the GaN substrate **101** to the growth rate of the vertical direction parallel to the 1-100 axis of the GaN substrate **101**. The ratio of ELO is preferably high, wherein the ratio of $\text{ELO} = \text{lateral growth rate} / \text{vertical growth rate}$. Optimizing the growth conditions, the ratio of ELO can be controlled from 0.4 to 4.

(121) Next, the III-nitride device layers **106** are grown on the ELO III-nitride layer **105**. The III-nitride device layers **106** are comprised of a plurality of III-nitride-based layers.

(122) Growth Restrict Mask

(123) The growth restrict mask **102** is disposed directly upon a substrate **101** that comprises a III-nitride based semiconductor. Specifically, the growth restrict mask **102** is disposed directly in contact, or is disposed indirectly through an intermediate layer grown by MOCVD or sputter, etc., made of a III-nitride based semiconductor. Two examples of the growth restrict mask **102** are shown in FIGS. 7(a) and 7(b).

(124) The growth restrict mask **102** shown in FIG. 7(a) comprises a plurality of opening areas **103**, which are arranged in a first direction parallel to the 11-20 direction of the (1-100) plane-oriented III-nitride semiconductor substrate **101** and a second direction parallel to the 0001 direction of the III-nitride semiconductor substrate **101**, periodically at intervals p_1 and p_2 , respectively, extending in the second direction. Each of a pair of the opening areas **103** are adjacent to each other in the second direction for the length q , respectively. The length a of the opening area **103** is, for example, 200 to 3500 μm ; the width b is, for example, 2 to 180 μm ; the interval p_1 of the opening areas **103** is, for example, 20 to 180 μm ; and the interval p_2 is, for example, 200 to 2000 μm .

(125) The island-like III-nitride semiconductor layers **109** have long sides and short sides.

Preferably, the long sides are perpendicular to the a -axis direction. In this case, as shown in the rightmost image of FIG. 6(b), the cores **601** of the epi-layer are aligned along perpendicular to the a -axis direction at the center of the opening area **103**. By doing this, it can control the beginning of growth, so that it is easy to obtain a smooth surface in the flat surface, region **107** of the ELO III-nitride layer **105**.

(126) Typical Dimensions of the Growth Restrict Mask

(127) Typically, the growth restrict mask **102** used in the present invention has dimensions as follows. In one embodiment, an m -plane GaN-substrate **101** is used. The growth restrict mask **102** is formed as shown in FIG. 7(c) with a 0.3 μm thick SiO_2 film, wherein the length a of the opening area **103** is 1200 μm ; the width b is 20 μm ; the interval p_1 of the opening area **103** is 80 μm ; the interval p_2 is 1300 μm ; the width of the mask **102** stripes between opening areas **103** is 60 μm ; and the distance between the opening area **103** and the opening area **103** in the 1-100 direction is 100 μm .

(128) Opening Areas of the Growth Restrict Mask

(129) The growth restrict mask **102** shown in FIG. 7(b) comprises a plurality of opening areas **103**, which are arranged in a first direction parallel to the 11-20 direction of the (1-100) plane-oriented III-nitride semiconductor substrate **101** periodically with the interval p_1 and extend in a second direction parallel to the 0001 direction of the III-nitride semiconductor substrate **101**. To prevent embossment of both end portions in the 0001 direction of the III-nitride semiconductor substrate **101** as described later, the growth restrict mask **102** further comprises a plurality of opening areas **103**, which are arranged periodically in the first direction at the same interval p_1 as the opening window **103**, shifted by a half of the interval p_1 with respect to the opening area **103**, and extend in the second direction in a manner such that the plurality of opening areas **103** overlap with the end portions of the opening areas **103** for the length q in the second direction.

(130) The length a of the opening area **103** is, for example, 200 to 3500 μm ; the width b is, for

example, 4 to 60 μm ; the interval $p1$ of the opening area **103** is, for example, 20 to 120 μm ; the width of the mask **102** stripes L is, for example, $p1-b$, so, in case of $p1=55\text{ }\mu\text{m}$ and $b=5\text{ }\mu\text{m}$, L is 50 μm ; and the overlapping length r of the end portions each other of the opening areas **103** is 35 to 40 μm .

(131) Advantages the Opening Areas

(132) Growing the ELO III-nitride layer **105** and island-like III-nitride semiconductor layers **109** using the growth restrict mask **102** shown in FIG. 1 provides many advantages.

(133) In the growth restrict mask **102** shown in FIGS. 4 and 7, as the longitudinal direction of the opening area **103** is in the 0001 direction, the growth rate of the III-nitride based semiconductor is low at the both ends of the opening area **103**, the island-like III-nitride semiconductor layers **109** opposing each other in the 0001 direction do not coalesce, and the island-like III-nitride semiconductor layers **109** can be separated from each other. At this time, the size of the island-like III-nitride semiconductor layer **109** in the 0001 direction becomes nearly equal with the length a of the opening area **103**.

(134) In the island-like III-nitride semiconductor layer **109** enclosed with planes, including planes with a low growth rate, when the distance is large between the adjacent III-nitride semiconductor layers **109** in opposing planes with a low growth rate, the following disadvantages occur. That is, in the stripes of the growth restrict mask **102** between adjacent island-like III-nitride semiconductor layers **109**, as the raw gas is not consumed there, the gas concentration goes up, and the concentration gradient is generated in the direction joining the adjacent III-nitride semiconductor layers **109**, and by the diffusion according to the concentration gradient, a lot of the raw gas is supplied at the edge portions of the island-like III-nitride semiconductor layers **109**. As the result, the thickness of the edge portions of the island-like III-nitride based semiconductor layers **109** becomes larger in comparison with other portions, and becomes a raised shape. More specifically, at the stripes of the growth restrict mask **102** between the adjacent island-like III-nitride based semiconductor layers **109** in the 0001 direction of which the growth rate is smaller, the raw gas is not consumed there, and therefore, the gas concentration goes up, and the concentration gradient in the 0001 direction is generated, and by the diffusion according to the concentration gradient, a lot of the raw gas is supplied at the edge portions in the 0001 direction of the island-like III-nitride semiconductor layers **109**. As the result, the thickness of the edge portions in the 0001 direction of the island-like III-nitride semiconductor layers **109** becomes larger in comparison with other portions, and becomes a raised shape. The specific raised shape of the edge portions of the island-like III-nitride semiconductor layers **109** causes not only structural inconveniences in the III-nitride based semiconductor device, but also creates problems in the following manufacturing processes of photolithography, etc.

(135) To prevent uniformity of the thickness of the island-like III-nitride semiconductor layer **109** caused by a specific raised shape of the edge portion, the adjacent island-like III-nitride semiconductor layers **109** come as close as possible, and it is necessary not to create in-plane uniformity of the raw gas from the beginning of the growth. For this, in the growth restrict mask **102** shown in FIG. 7(b), upon the bisecting line of the region between a pair of opening areas **103** adjacent to each other in the 11-20 direction, is formed in a manner such that the opening area **103** overlaps with the opposing end portions of the opening area **103** adjacent to each other in the 0001 directions for the length q , respectively.

(136) As a result, in-plane uniformity of gas concentration can be obtained by the consumption of the raw gas caused by growing the island-like III-nitride semiconductor layers **109**. Finally, a uniformity in the thickness of the island-like III-nitride semiconductor layers **109** can be obtained.

(137) Growth Conditions of the ELO III-Nitride Layer

(138) The growth conditions of the island-like III-nitride semiconductor layer **109** can be the same MOCVD growth conditions as the ELO III-nitride layers **105**. For example, the growth of the GaN layer is at a temperature of 930-1200° C. and a pressure of 15 kPa. For the growth of a GaN layer,

trimethylgallium (TMG) and ammonia (NH₃) are used as the raw gas, and the carrier gas is only Hydrogen (H₂), with silane (SiH₄) as a dopant gas. The growth time is 4 hours.

(139) The growth gas flow rate is following; TMG is 12 sccm, NH₃ is 8 slm, carrier gas is 3 slm, and SiH₄ is 1.0 sccm. The V/III ratio is about 7700. Then, it can obtain a 20 μm thick ELO III-nitride layer **105**.

(140) Effects of Isolation of a Region from Growth

(141) Also, as shown in FIGS. 4(a)-4(b), the III-nitride based semiconductor layers are isolated from the edge of the substrate **101**.

(142) FIG. 10(a) is a photograph of a GaN layer on a semi-polar (20-21) substrate without a pattern, wherein the shapes of the edges of the substrate **101**, which are enclosed by dashed lines, are not uniform. Consequently, the growth rate and the shape of the core is unstable, so that the edge region of the substrate **101** has a rough surface.

(143) A non-polar III-nitride substrate **101**, such as an m-plane III-nitride substrate **101**, has similar results. As shown in the two images on the left side of FIG. 10(b), there are a lot of cores **1001** of random shape on the substrate **101** with no pattern, which results in a large amount of surface roughness. On the other hand, there is a smooth surface on the substrate **101** patterned with a growth restrict mask **102** of SiO₂, due to the isolated areas, as shown in the two images on the right side of FIG. 10(b).

(144) At least the opening area **103** should be isolated from the facets of the substrate **101**, which are enclosed by dashed lines in FIG. 10(a). These facets are perpendicular to the a-axis. Preferably, the opening area **103** is isolated from the facets that are perpendicular from the projected c-axis.

(145) Consequently, to obtain a smooth surface in the flat surface region **107**, the III-nitride based semiconductor layers should be isolated from the edge of substrate **101**.

(146) Other examples are shown in the images of FIGS. 10(c)-10(d), wherein the substrate of FIG. 10(c) is (10-1-1) and the substrate of FIG. 10(d) is (1-100) without an off-angle orientation. Both FIGS. 10(c) and 10(d) include images of substrates **101** that are not patterned **1002** and patterned **1003**, that were grown with MOCVD at the same time. The surface morphology of the substrates **101** that are not patterned **1002** is rough, with non-uniform edges and pits on the surface. On the other hand, the substrates **101** that are patterned **1003** have a very smooth surface. The enlarged portions of the substrates **101** that are patterned **1003** show some debris on the growth restrict mask **102**, but the surface of the ELO III-nitride layer **105** is very smooth. Moreover, the debris disappears when the growth conditions are optimized. Generally, a (10-1-1) plane is likely to obtain a rough surface, but the use of a substrate **101** that is patterned **1003** can result in a smooth surface.

(147) Different Carrier Gas Conditions

(148) To compare the effects depending on carrier gas, two samples have been grown with a Hydrogen carrier gas condition and a Nitrogen carrier gas condition.

(149) As shown in the three images of FIG. 6(a), which reflect growth on the (1-100), (20-21) and (20-2-1) planes, the Nitrogen carrier gas condition results in a depressed region on the surface at the center of ELO III-nitride layer **105**. As mentioned above, the reason is the lack of Hydrogen etching at the edges of the opening area **103**.

(150) As shown in the two images of FIG. 6(c), the Hydrogen carrier gas condition does not have the same problem.

(151) As shown in the two images of FIG. 6(b), the cores **601** made at the beginning of the growth are formed at the center of the opening region **103**. This growth has avoided coalescing the cores **601** at the center, so that the depressed region has disappeared from the ELO III-nitride layer **105**.

(152) FIGS. 6(a), 6(b) and 6(c) are results in the case where the width of the opening area **103** is 8-9 μm.

(153) On the other hand, the twelve images of FIG. 6(h) show the results in the cases where the width of the opening area **103** is 25 μm. In these cases also, using a Hydrogen carrier gas, alone or in combination with a Nitrogen carrier gas, resulted in a smooth surface, but using a Nitrogen

carrier gas case alone resulted in the surfaces having a large amount of surface roughness without flat areas.

(154) Furthermore, the region of growth has been isolated from the edge of the substrate **101**, which keeps hillocks from appearing on the surface.

(155) Different Off-Angle Orientations

(156) Different off-angle orientations of the substrate **101** have been prepared, along with a just-orientation. The off-angle orientations are of an m-plane oriented crystalline surface plane, wherein the off-angle orientations range from about +47 degrees to about -47 degrees towards a c-plane. ELO III-nitride layers **105** were grown on those substrates **101**, with the Hydrogen carrier gas condition, at the same time.

(157) FIG. 6(d) includes four images of the surface of the ELO III-nitride layer **105**, where the angle of the substrates **101**, from left to right, is 0, -0.45, -0.6 and -1 degrees from the m-plane towards the (0001) c-plane. It would be expected that these off-angle orientations would affect surface morphology, which would be the case when using a substrate **101** with no pattern. Some substrates **101** would have pyramidal hillocks, wavy surfaces, etc.

(158) However, as shown in FIG. 6(d), every sample has a good surface morphology at the same time, even with different off-angle orientations. In m-plane growth, such results have not been reported before.

(159) Furthermore, as shown in the eight images of FIG. 6(f), smooth surfaces can be realized with the same technique on substrates **101** with different planes, such as (10-10), (20-2-1), (10-1-1), (10-1-2), (20-21), (10-11) and (10-12), that are on-axis (0 degrees off-angle orientation) or mis-oriented (-15, -28, -47, +15, +28, +47 degrees off-angle orientation), using only H.sub.2 carrier gas.

(160) These samples have a very smooth surface. The (10-1-2) and (10-11) planes have a surface that is a little bit rough, but a part of the surface shows a flat area. An optimization of the growth conditions, such as the ratio of Hydrogen and Nitrogen of the carrier gas, V/III ratio, and growth temperature and so on, would improve the surface roughness. Therefore, the present invention can also adopt these planes. On the other hand, the (10-12) plane has a triangular shape without a flat area.

(161) As shown in the images of FIG. 6(i), the present invention can adopt these substrates **101** which have an off angle orientation ranging from -47 degrees to +28 degrees from the m-plane towards the c-plane. More preferably, the off-angle orientation ranges from -28 degrees to +15 degrees from the m-plane towards the c-plane.

(162) In addition, growth may be performed with a mixed gas condition using a Hydrogen and Nitrogen carrier gas. Both gas flows are 1.5 slm and total carrier gas flow is 3.0 slm.

(163) The results using an H.sub.2 and N.sub.2 mixed carrier gas are shown in the images of FIG. 6(g), as well as FIG. 6(e). In these cases, substantially the same results are obtained for the (10-10), (20-2-1), (10-1-1), (10-10), and (20-21) surfaces, at 0, -15, -28, 0 and +15 degrees, respectively. Thus, including Hydrogen in the carrier gas is important for obtaining a smooth surface.

(164) Consequently, a very smooth surface can be obtained, even though a different off-angle orientation and plane is used. This is especially true for off-angles under 0.6 degrees, where it has been difficult to obtain smooth surfaces. However, even for off-angles under 0.6 degrees, it is possible to obtain a smooth surface using the present invention.

(165) Generally, a GaN substrate **101** has a large in-plane distribution of off-angle orientations. Previously, though, the large in-plane distribution on off-angle orientations resulted in a decreasing yield due to rough surfaces, which is a major problem.

(166) As shown in FIGS. 11(a)-11(b), the off-angle orientation is different at each point A, B and C. Consider an example where point A is an m-plane without an off-angle orientation, i.e., on-axis, point B has an off-angle orientation of 0.1 degrees from the m-plane, and point C has an off-angle orientation of 0.2 degrees from the m-plane. In this example, surface morphology is different at each point A, B and C.

(167) The present invention, on the other hand, can avoid this problem and obtain a smooth surface at every point A, B and C, making mass production more efficient and lower cost.

(168) Thus, in cases where the substrate **101** has a fluctuation of the in-plane distribution of off-angle orientations with more than 0.1 degrees, and more preferably more than 0.2 degrees, the present invention is very useful.

(169) III-Nitride Semiconductor Device Layers

(170) As a next step, III-nitride semiconductor device layers **106** have been grown on ELO III-nitride layers **105**. For the growth of an AlGaN layer, triethylaluminium (TMA) is used as the raw gas; and for the growth of an InGaN layer, trimethylindium (TMI) is used as the raw gas. Under these conditions, the following layers have been grown on ELO III-nitride layers **105**.

(171) FIG. **12** is a cross sectional view of a nitride semiconductor laser bar along the direction perpendicular to an optical resonator, wherein the optical resonator is comprised of a ridge stripe structure.

(172) The nitride semiconductor laser has the following layers, laid one on top of another in the order mentioned, an ELO III-nitride layer **105** (which is a GaN layer), an InGaN/GaN 5 MQW active layer **1201** (8 nm×8 nm: 5 MQW), an AlGaN-EBL (electron blocking layer) layer **1202**, a p-GaN guiding layer **1203**, a ZrO.sub.2 current limiting layer **1204**, and a p-electrode **1205**. Note that these III-nitride semiconductor layers may be formed of any nitride-based III-V group compound semiconductor grown in the above order.

(173) The ridge stripe structure is comprised of the p-GaN cladding layer **1203**, ZrO.sub.2 current limiting layer **1204**, and p-electrode **1205**, provides optical confinement in a horizontal direction. The width of the ridge stripe structure is of the order of 1.0 to 20 μm , and typically is 10 μm .

(174) In one embodiment, the p-electrode **1205** may be comprised of one or more of the following materials: Pd, Ni, Ti, Pt, Mo, W, Ag, Au, etc. For example, the p-electrode **1205** may comprise Pd-Ni-Au (with thicknesses of 3-30-300 nm). These materials may be deposited by electron beam evaporation, sputter, thermal heat evaporation, etc.

(175) In addition, an ITO cladding layer may be added between p-GaN cladding layer **1203** and p-electrode **1205**, as shown in FIG. 2(c).

(176) Initial Growth

(177) To obtain a smooth surface with non-polar or semi-polar substrates **101**, it is necessary to consider the mechanism of the initial growth, which is different from polar c-plane substrates. As shown in the images of FIG. **13**, the initial growth on a polar c-plane substrate **101** with a (0001) surface shows an isotropic growth such as hexagonal shapes **1301**. On the other hand, the initial growth of a non-polar substrate **101** that has an off-angle orientation tilting from the m-plane (10-10) to the c-plane, shows an anisotropic growth, also as shown in FIG. **13**.

(178) This phenomenon has been explained by some researchers. For example, Lymperakis and Neugebauer have calculated the highly anisotropic diffusion barriers for Ga adatoms on m-plane GaN surfaces along the c- and a-directions as 0.93 eV and 0.21 eV, respectively. See Phys. Rev. B 79, 241308(R). It is thought that the anisotropic diffusion barriers for Ga adatoms cause the anisotropic initial growth. Besides, this anisotropic initial growth causes a large amount of surface roughness in the case where there is no limitation of the growth area, as shown for the initial growth of m-plane without a mask as shown in FIG. **14(a)** and with a mask **102** as shown in FIG. **14(b)**.

(179) For the above reasons, non-polar substrates **101** having an off-angle orientation tilting from the m-plane to the c-plane grow rapidly at the beginning of the growth in the long dimension along the a-axis rather than the c-axis. In this case, the growth restrict mask **102** can form openings **103** with long sides in the a-direction rather than projected with the c-axis, which makes the surface smooth due to being able to control the initial growth position and avoiding the coalescing of the initial growth from unintentional directions, as shown in FIGS. **16** and **17**. A non-polar substrate **101** having an off-angle orientation of the m-plane oriented crystalline surface plane ranges from

about +28 degrees to about -47 degrees towards a c-plane.

(180) For the above reasons, these substrates might obtain a smooth surface using the growth restrict mask **102**.

(181) FIG. **13** also shows similar results to the non-polar (10-10) substrate **101** for semi-polar (20-21) and (20-2-1) substrates **101**.

(182) Edge Growth

(183) The ELO III-nitride layers **105** have a just-orientation and an off-angle orientation from the m-plane oriented crystalline surface plane, wherein the off-angle orientation ranges from about +28 degrees to about -47 degrees toward a c-plane. The III-nitride semiconductor device layers **106** are grown on the ELO III-nitride layers **105**. In this case, the edge growth has been restricted.

(184) As shown in FIGS. **15(a)**-**15(c)**, in the case of a c-plane substrate **101**, there are the edge growth regions **1501** at the edges of the island-like III-nitride semiconductor layers **109**. Generally, the width W of the edge growth regions **1501** is about 10-15 μm and the height T of the edge growth region **1501** is 0.3-0.4 μm . The edge growth region **1501** has a high growth rate, so that each layer thickness is different from at the center of the island-like III-nitride semiconductor layers **109**.

(185) In this case, it would be better to avoid forming the ridge stripe structure from the edge growth regions **1501**, because the production yield would decrease due to the fluctuations of the thickness of the layers. In the case of the c-plane substrate **101**, the edge growth region **1501** is wide and high. However, in the present invention, the height T is less than 0.2 μm , and the width W is about 5 μm .

(186) Limiting the edge growth region **1501** is very important in the fabrication of the device. By optimizing growth conditions, the edge growth region **1501** can substantially disappear.

(187) Making the Ridge Stripe Structure

(188) Using conventional methods, such as photo-lithography and dry etching, a ridge stripe structure **301** is fabricated, as shown in FIGS. **3(a)**-**3(b)**, after MOCVD growth. The ridge depth (from the surface to the ridge bottom) is in the p-GaN guide layer. The ridge depth is pre-determined before dry etching is performed, based on simulation or previous experimental data.

(189) Method of Making a Facet

(190) As shown in FIGS. **3(a)**-**3(b)**, the etched mirror region **302** is located based on optical resonance length. The etching process for GaN etching uses an Ar ion beam and Cl.sub.2 ambient gas. The etching depth is from about 1 μm to about 4 μm . The etched mirror facet may be coated by a dielectric film selected from the group of the following: SiO.sub.2, Al.sub.2O.sub.3, AlN, AlON, SiN, SiON, TiO.sub.2, Ta.sub.2O.sub.5, Nb.sub.2O.sub.5, Zr.sub.2O, etc.

(191) The III-Nitride Based Device Layers are Bonded to the Support Substrate

(192) FIG. **2(d)** shows the first support substrate **204**, which is bonded to the island-like III-nitride semiconductor layers **109**. Conventional bonding techniques can be adopted for the bonding.

(193) In general, the most common types of flip-chip bonding are thermal compression bonding and wafer fusion/bonding. Wafer fusion has been popularly employed in InP-based devices. However, thermal compression bonding is generally much simpler than wafer fusion, as it uses metal-to-metal bonding, and has the benefit of also greatly improving thermal conductivity.

(194) An Au—Au compression bond is by far the simplest bond and results in a fairly strong bond. An Au—Sn eutectic bond offers a much greater bond strength.

(195) In one embodiment, a Cu substrate is used as the support substrate **204**. A patterned Ti/Au electrode is fabricated on the Cu substrate by electron beam evaporation or sputter. The thickness of the electrode is Ti (10 nm), and Au (500 nm).

(196) It is preferably to perform an activation of the surface for wafer bonding before compression bonding. The activation of the surface is achieved by using a plasma process of Ar and/or O.sub.2. The island-like III-nitride semiconductor layers **109** are then bonded to the support substrate **204** at 150-300° C. under pressure.

(197) Removing the Substrate by Thermal Expansion

(198) The bonded wafer is dipped into a solvent for wet etching to remove the substrate **101**. In one embodiment, the growth restrict mask **102** used is SiO₂, which is dissolved by an HF or BHF solvent. The merit of this technique is that there is no mechanical damage when the substrate **101** is removed (very gently), and a wide area of SiO₂ is dissolved by the HF very easily and quickly.

(199) Thereafter, the wafer, which is bonded to the III-nitride substrate **101** and support substrate **204**, is heated. A Cu support substrate **204** has a larger CTE (Coefficient of Thermal Expansion) than a GaN substrate **101**. As shown FIG. 2(e), the expansion of the support substrate **204** due to heating applies strong stress to a cleaving point **205**. Afterward, the cleaving starts at the cleaving point **205** towards an opposite side of a cleaving surface **206**.

(200) N-Electrode

(201) As noted in FIG. 2(f), an n-electrode **207** is placed on the back side of the island-like III-nitride semiconductor layers **109**. Typically, the n-electrode **207** is comprised of the following materials: Ti, Hf, Cr, Al, Mo, W, Pt, Au.

(202) For example, the n-electrode **207** may be comprised of Ti—Al—Pt—Au (with a thickness of 30-100-30-500 nm), but is not limited to those materials. The deposition of these materials may be performed by electron beam evaporation, sputter, thermal heat evaporation, etc. Preferably, the p-electrode **207** is deposited on the ITO.

(203) As shown in FIG. 16, the n-electrode (not shown) preferably is formed in a separate area **1601** on the back-side of the island-like III-nitride semiconductor layers **109**, after the island-like III-nitride semiconductor layers **109** are removed from the substrate **101**. This separate area **1601** has a good surface condition for the n-electrode to obtain low contact resistivity. Specifically, the present invention keeps this separate area **1601** clean until the island-like III-nitride semiconductor layers **109** are removed.

(204) Chip Division Method

(205) The chip division method has two steps. The first step is to scribe the island-like III-nitride semiconductor layers **109**. The second step is to divide the support substrate **204** using a laser scribe, etc.

(206) As shown in FIGS. 3(a)-3(b), the chip scribe line **303** is fabricated by a diamond scribing machine or laser scribe machine. The chip scribe line **303** is fabricated on the back side of the island-like III-nitride semiconductor layers **109**. The chip scribe line **303** may be a solid line or a dashed line.

(207) Next, the support substrate **204** is divided by laser scribing as well to obtain an LD device. It is better to avoid the ridge strip structure when the chip scribe line **303** is fabricated.

Second Embodiment

(208) A second embodiment is almost the same as the first embodiment, except that it does not remove the island-like III-nitride semiconductor layers **109**. The steps of the fabrication method are the same as the first embodiment, until step 3 (TCO p-pad deposition+ridge process). In the second embodiment, the island-like III-nitride semiconductor layers **109** do not bond to the support substrate **204**. The following process is the same of conventional device process.

(209) 4. Polishing the substrate **101** until it is 80-100 μm in thickness.

(210) 5. Forming an n-electrode on the back side of the substrate **101**.

(211) 6. Separating the substrate **101** into the bars, as shown in FIGS. 17(a)-17(b), wherein FIG. 17(a) shows the flat surface regions **107**, layer bending regions **108**, and island-like III-nitride based semiconductor layers **109**, separated by no-growth regions **104**, as well as scribe lines **1701**; and FIG. 17(b) shows the devices separated along the scribe lines **1701** into bars **1702**.

(212) 7. Facet coating (using the same method as mentioned in the first embodiment).

(213) 8. Separating the bars **1702** into individual devices or chips **1703**, as shown in FIGS. 17(b)-17(c).

(214) By doing this, the devices **1703** can be obtained without removing the island-like III-nitride

semiconductor layers **109** from the substrate **101**. By using the method of this second embodiment, the same effect as the first embodiment is obtained.

(215) Modifications and Alternatives

(216) A number of modifications and alternatives can be made without departing from the scope of the present invention.

(217) For example, the present invention may be used with III-nitride substrates of other orientations. Specifically, the substrates may be basal non-polar m-plane $\{1\ 0\ -1\ 0\}$ families; and semi-polar plane families that have at least two nonzero h, i, or k Miller indices and a nonzero 1 Miller index, such as the $\{2\ 0\ -2\ -1\}$ planes. Semi-polar substrates of (20-2-1) are especially useful, because of the wide area of flattened ELO growth.

(218) In another example, the present invention is described as being used to fabricate different opto-electronic device structures, such as a light-emitting diode (LED), laser diode (LD), Schottky barrier diode (SBD), or metal-oxide-semiconductor field-effect-transistor (MOSFET). The present invention may also be used to fabricate other opto-electronic devices, such as micro-LEDs, vertical cavity surface emitting lasers (VCSELs), edge-emitting laser diodes (EELDs), and solar cells.

(219) Conclusion

(220) This concludes the description of the preferred embodiment of the present invention. The foregoing description of one or more embodiments of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention be limited not by this detailed description, but rather by the claims appended hereto.

Claims

1. A device, comprising: one or more island-like III-nitride semiconductor layers having a just-orientation and an off-angle orientation of an m-plane oriented crystalline surface plane, wherein: the island-like III-nitride semiconductor layers are grown in opening areas in a growth restrict mask deposited on a substrate; the just-orientation of the m-plane oriented crystalline surface plane of the island-like III-nitride semiconductor layers is towards an a-axis of the island-like III-nitride semiconductor layers; the off-angle orientation of the m-plane oriented crystalline surface plane of the island-like III-nitride semiconductor layers ranges from about +28 degrees to about -47 degrees towards a c-plane of the island-like III-nitride semiconductor layers; the island-like III-nitride semiconductor layers have at least one long side and short side, wherein the long side is perpendicular to an a-axis of the island-like III-nitride semiconductor layers; the island-like III-nitride semiconductor layers have an edge growth region at the short side of the island-like III-nitride semiconductor layers where growth of the island-like III-nitride semiconductor layers is restricted with a height less than 0.2 μm ; and the island-like III-nitride semiconductor layers have a flat surface region without hillocks.
2. The device of claim 1, wherein the growth restrict mask is isolated from an edge of the III-nitride substrate.
3. The device of claim 2, wherein the island-like III-nitride semiconductor layers are removed from the III-nitride substrate.
4. The device of claim 1, wherein the island-like III-nitride semiconductor layers have an emitting region.
5. The device of claim 4, wherein the emitting region is at least 1 μm from an edge of a layer bending region of the island-like III-nitride semiconductor layers, and the layer bending region is a region including a bended active region.
6. The device of claim 4, wherein the emitting region is more than 5 μm from an edge of a top surface of the island-like III-nitride semiconductor layers.

7. The device of claim 1, wherein the edge growth region has a width less than 5 μm .
 8. The device of claim 1, wherein the island-like III-nitride semiconductor layers have a separate region where an n-electrode is formed.
 9. A method of fabricating a semiconductor device, comprising: forming a growth restrict mask on or above a III-nitride substrate, wherein: the III-nitride substrate has an in-plane distribution of off-angle orientations with more than 0.1 degrees; and the off-angle orientations of an m-plane oriented crystalline surface plane ranges from about +28 degrees to about -47 degrees towards a c-plane; and growing one or more island-like III-nitride semiconductor layers on the III-nitride substrate using the growth restrict mask; wherein: the island-like III-nitride semiconductor layers have a just-orientation of an m-plane oriented crystalline surface plane towards an a-axis; the island-like III-nitride semiconductor layers have the off-angle orientations of the m-plane oriented crystalline surface plane ranges from about +28 degrees to about -47 degrees towards a c-plane; the island-like III-nitride semiconductor layers have at least one long side and short side, wherein the long side is perpendicular to an a-axis of the island-like III-nitride semiconductor layers; the island-like III-nitride semiconductor layers have an edge growth region at the short side of the island-like III-nitride semiconductor layers where growth of the island-like III-nitride semiconductor layers is restricted with a height less than 0.2 μm ; the island-like III-nitride semiconductor layers have a flat surface region without hillocks.
 10. The method of claim 9, wherein the island-like III-nitride semiconductor layers are removed from the III-nitride substrate.
 11. The method of claim 9, wherein the island-like III-nitride semiconductor layers have an emitting region.
 12. The method of claim 11, wherein the emitting region is at least 1 μm from an edge of a layer bending region of the island-like III-nitride semiconductor layers, and the layer bending region is a region including a bended active region.
 13. The method of claim 11, wherein the emitting region is more than 5 μm from an edge of a top surface of the island-like III-nitride semiconductor layers.
 14. The method of claim 9, wherein the edge growth region has a width less than 5 μm .
 15. The method of claim 9, wherein the island-like III-nitride semiconductor layers have a separate region where an n-electrode is formed.
 16. The method of claim 9, wherein the growth restrict mask is isolated from an edge of the III-nitride substrate.
 17. The method of claim 9, wherein the island-like III-nitride semiconductor layers do not coalesce with neighboring island-like III-nitride semiconductor layers, resulting in a no-growth region on the growth restrict mask between the island-like III-nitride semiconductor layers and the neighboring island-like III-nitride semiconductor layers.
 18. The device of claim 1, wherein the island-like III-nitride semiconductor layers do not coalesce with neighboring island-like III-nitride semiconductor layers, resulting in a no-growth region on the growth restrict mask between the island-like III-nitride semiconductor layers and the neighboring island-like III-nitride semiconductor layers.
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